Loading lab data: first exploration

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In this section, we outline the initial steps taken to load and inspect the lab data set. The primary goal is to understand the structure of the data and to translate column names from German to English.

1.1 Load the data and necessary packages

```r
# Load Packages
library(data.table)

# Input file path for the lab data
lab_data_path <- "./RITM0154633_del_20240923/RITM0154633_lab.csv"
lab_data <- read.csv(lab_data_path)
```

1.2 Basic data inspection

- Check the first rows to see the structure of the data set:

```r
head(lab_data)
```

```
##   dim_patient_bk_pseudo dim_fall_bk_pseudo bezeichnung kurzbezeichnung
## 1                     1             171465     Natrium            <NA>
## 2                     1             171465      Kalium              KA
## 3                     1             171465 Hämolytisch            H-Se
## 4                     1             171465   Lipämisch            L-Se
## 5                     1             171465   Ikterisch            I-Se
## 6                     1             171465 Hämolytisch               H
##   methodenummer ergebnis_numerisch ergebnis_text    unit
## 1             1                138           138  mmol/L
## 2             3                4.6           4.6  mmol/L
## 3            42                  4             4 Unknown
## 4            43                  3             3 Unknown
## 5            44                  1             1 Unknown
## 6            47                 10            10 Unknown
```

The data set contains columns such as patient and case pseudonyms, test names, abbreviations, and numerical results.

- Print column names:

```r
names(lab_data)
```

```
## [1] "dim_patient_bk_pseudo" "dim_fall_bk_pseudo"    "bezeichnung"
## [4] "kurzbezeichnung"       "methodenummer"         "ergebnis_numerisch"
## [7] "ergebnis_text"         "unit"
```

- Since the data set contains German terms, translate columns for non-German speakers:

```r
colnames(lab_data) <- c("pseudo_patient_id", "pseudo_case_id", "test_name",
                        "test_abbr", "method_num", "num_result",
                        "text_result", "unit")
```

1.3 Data dimensions and overview

- Get the dimensions, column names (after translation), structure and summary statistics:

```r
# Number of rows and columns
dim(lab_data)
```

```
## [1] 20487004        8
```

The lab_data data set consists of 20,487,004 observations and 8 variables.

```r
# Column names
names(lab_data)
```

```
## [1] "pseudo_patient_id" "pseudo_case_id"    "test_name"
## [4] "test_abbr"         "method_num"        "num_result"
## [7] "text_result"       "unit"
```

```r
# View structure of the dataset
str(lab_data)
```

```
## 'data.frame':    20487004 obs. of  8 variables:
##  $ pseudo_patient_id: int  1 1 1 1 1 1 1 1 1 1 ...
##  $ pseudo_case_id   : int  171465 171465 171465 171465 171465 171465 171465 171465 171465 171465 ...
##  $ test_name        : chr  "Natrium" "Kalium" "Hämolytisch" "Lipämisch" ...
##  $ test_abbr        : chr  NA "KA" "H-Se" "L-Se" ...
##  $ method_num       : int  1 3 42 43 44 47 47 48 48 49 ...
##  $ num_result       : chr  "138" "4.6" "4" "3" ...
##  $ text_result      : chr  "138" "4.6" "4" "3" ...
##  $ unit             : chr  "mmol/L" "mmol/L" "Unknown" "Unknown" ...
```

```r
# Get summary statistics for all columns
summary(lab_data)
```

```
##  pseudo_patient_id pseudo_case_id    test_name          test_abbr
##  Min.   :     1    Min.   :     1   Length:20487004    Length:20487004
##  1st Qu.: 64105    1st Qu.:110462   Class :character   Class :character
```

```
##  Median :127885    Median :221988   Mode  :character   Mode  :character
##  Mean   :127043    Mean   :220664
##  3rd Qu.:192049    3rd Qu.:332397
##  Max.   :240990    Max.   :428141
##    method_num     num_result        text_result            unit
##  Min.   :    1   Length:20487004    Length:20487004    Length:20487004
##  1st Qu.:   51   Class :character   Class :character   Class :character
##  Median :  551   Mode  :character   Mode  :character   Mode  :character
##  Mean   : 5565
##  3rd Qu.: 9111
##  Max.   :95999
```

1.4 Additional data checks

- Check for missing values:

```r
missing_values <- sapply(lab_data, function(x) sum(is.na(x)))
print(missing_values)
```

```
## pseudo_patient_id    pseudo_case_id         test_name         test_abbr
##                 0                 0                 0            387136
##        method_num        num_result       text_result              unit
##                 0                 0                39                 0
```

*It has **0** missing values for most columns, but the `test_abbr` column has **387,136** missing entries, and the `text_result` column has **39** missing values.*

- Get unique values for all columns and count them:

```r
unique_values_list <- lapply(lab_data, unique)
unique_counts <- sapply(unique_values_list, length)
unique_df <- data.frame(
  Unique_Count = unique_counts,
  Unique_Values = I(unique_values_list)  # Using I() to prevent auto-unlisting
)
print(unique_df)
```

```
##                   Unique_Count Unique_Values
## pseudo_patient_id       182238  1, 2, 3,....
## pseudo_case_id          311611  171465, ....
## test_name                 3470  Natrium,....
## test_abbr                 4609  NA, KA, ....
## method_num                4695  1, 3, 42....
## num_result               62306  138, 4.6....
## text_result              98460  138, 4.6....
## unit                       169  mmol/L, ....
```

- Get unique test abbreviations, sort them, and display the result:

```r
unique_test_abbr <- sort(unique(lab_data$test_abbr))
unique_abbr_df <- data.frame(Unique_Test_Abbreviations = unique_test_abbr)
head(unique_abbr_df)
```

```
##   Unique_Test_Abbreviations
## 1                     #BASO
## 2                      #BZM
## 3                    #BZMns
## 4                    #BZMsv
## 5                     #BZNa
## 6                      #DC1
```

- Get unique test names, sort them, and display the result:

```r
unique_test_names <- sort(unique(lab_data$test_name))

unique_names_df <- data.frame(Unique_Test_Names = unique_test_names)

head(unique_names_df)
```

```
##   Unique_Test_Names
## 1
## 2               -13
## 3             -13-2
## 4                -5
## 5              -5-2
## 6               -5q
```

- Create a list of unique pairs of test names and their abbreviations from the `lab_data` data frame. These pairs are then sorted by the `test_name` for easier viewing and analysis. Finally, look at the first few rows to verify that it's correct:

```r
# Create unique test_name and test_abbr pairs
# Get unique pairs of test_name and test_abbr
unique_test_pairs <- unique(lab_data[, c("test_name", "test_abbr")])

# Sort the pairs by test_name for better readability
unique_test_pairs_sorted <- unique_test_pairs[order(unique_test_pairs$test_name), ]

# View the first few rows of the sorted unique test pairs
head(unique_test_pairs_sorted)
```

```
##        test_name test_abbr
## 63169               B1adn+
## 63170               K2adn-
## 63171               B2adn-
## 147628              K1adn+
## 192266              B1adn-
## 233261              B1-dn+
```

- Check for duplicate rows in the data set:

```r
lab_data_dt <- as.data.table(lab_data) # Convert to data.table

# Find the duplicate rows
duplicates <- lab_data_dt[duplicated(lab_data_dt) | duplicated(lab_data_dt, fromLast = TRUE)]

# Count the number of duplicates
duplicates_count <- nrow(duplicates)
print(paste("Number of duplicate rows:", duplicates_count))
```

```
## [1] "Number of duplicate rows: 50"
```

```r
# Print the duplicate rows
print(duplicates)
```

```
##     pseudo_patient_id pseudo_case_id                test_name test_abbr
##                 <int>          <int>                   <char>    <char>
##  1:              2019          30187        Heparin, unfrakt.     HAXaU
##  2:              2019          30187        Heparin, unfrakt.     HAXaU
##  3:              6628         195679                Basophile    BASO2n
##  4:              6628         195679                Basophile    BASO2n
##  5:              8904         370137                      INR     INRiH
##  6:              8904         370137                      INR     INRiH
##  7:              9644          78608                Monozyten    MONO2n
##  8:              9644          78608                Monozyten    MONO2n
##  9:             14122         126261                      INR     INRiH
## 10:             14122         126261                      INR     INRiH
## 11:             19085         231531               Hämatokrit      Hk.n
## 12:             19085         231531               Hämatokrit      Hk.n
## 13:             19085         231531               Hämatokrit       Hkn
## 14:             19085         231531               Hämatokrit       Hkn
## 15:             22437         220248                      INR     INRiH
## 16:             22437         220248                      INR     INRiH
## 17:             49820         380729                      INR     INRiH
## 18:             49820         380729                      INR     INRiH
## 19:             51845         387034                      INR     INRiH
## 20:             51845         387034                      INR     INRiH
## 21:             55972         275993                      INR     INRiH
## 22:             55972         275993                      INR     INRiH
## 23:             95387         157179               Myelozyten       MY2
## 24:             95387         157179               Myelozyten       MY2
## 25:            101830         159671                      INR     INRiH
## 26:            101830         159671                      INR     INRiH
## 27:            111216          44091                      INR     INRiH
## 28:            111216          44091                      INR     INRiH
## 29:            123384         370703 Neutrophile Granulozyten    NEUTv#
## 30:            123384         370703 Neutrophile Granulozyten    NEUTv#
## 31:            143212         353396        Segmentkernige N.    SEGM2n
## 32:            143212         353396        Segmentkernige N.    SEGM2n
## 33:            156058         129237                      INR     INRiH
## 34:            156058         129237                      INR     INRiH
## 35:            161160         283661           Stabkernige N.    STAB2n
## 36:            161160         283661           Stabkernige N.    STAB2n
```

```
## 37:            161886         213149                      INR     INRiH
## 38:            161886         213149                      INR     INRiH
## 39:            165037          63856                      INR     INRiH
## 40:            165037          63856                      INR     INRiH
## 41:            179560          95473        Atyp. Lymphozyten    ALYM2n
## 42:            179560          95473        Atyp. Lymphozyten    ALYM2n
## 43:            183201         228469       Quot. %Mikro/%Hypo    MiHypo
## 44:            183201         228469       Quot. %Mikro/%Hypo    MiHypo
## 45:            206584           9895  Calcium Alb.-korrigiert       Cak
## 46:            206584           9895  Calcium Alb.-korrigiert       Cak
## 47:            212792         387238                      INR     INRiH
## 48:            212792         387238                      INR     INRiH
## 49:            221148         187319                      INR     INRiH
## 50:            221148         187319                      INR     INRiH
##     pseudo_patient_id pseudo_case_id                test_name test_abbr
##     method_num num_result text_result      unit
##          <int>     <char>      <char>    <char>
##  1:       6152       0.09       < 0.1 AntiXa/mL
##  2:       6152       0.09       < 0.1 AntiXa/mL
##  3:       9130       0.06        0.06       G/L
##  4:       9130       0.06        0.06       G/L
##  5:       6608       0.96        <1.0   Unknown
##  6:       6608       0.96        <1.0   Unknown
##  7:       9132       0.85        0.85       G/L
##  8:       9132       0.85        0.85       G/L
##  9:       6608       0.95        <1.0   Unknown
## 10:       6608       0.95        <1.0   Unknown
## 11:       9110       0.37        0.37       L/L
## 12:       9110       0.37        0.37       L/L
## 13:       9111       0.37        0.37       L/L
## 14:       9111       0.37        0.37       L/L
## 15:       6608       0.97        <1.0   Unknown
## 16:       6608       0.97        <1.0   Unknown
## 17:       6608       0.98        <1.0   Unknown
## 18:       6608       0.98        <1.0   Unknown
## 19:       6608       0.96        <1.0   Unknown
## 20:       6608       0.96        <1.0   Unknown
## 21:       6608       0.96        <1.0   Unknown
## 22:       6608       0.96        <1.0   Unknown
## 23:        142       0.08        0.08   *10^9/l
## 24:        142       0.08        0.08   *10^9/l
## 25:       6608       0.98        <1.0   Unknown
## 26:       6608       0.98        <1.0   Unknown
## 27:       6608       0.96        <1.0   Unknown
## 28:       6608       0.96        <1.0   Unknown
## 29:       9223       0.05        0.05       G/L
## 30:       9223       0.05        0.05       G/L
## 31:       9126       0.86        0.86       G/L
## 32:       9126       0.86        0.86       G/L
## 33:       6608       0.99        <1.0   Unknown
## 34:       6608       0.99        <1.0   Unknown
## 35:       9124       0.58        0.58       G/L
## 36:       9124       0.58        0.58       G/L
## 37:       6608       0.94        <1.0   Unknown
```

```
## 38:       6608       0.94        <1.0   Unknown
## 39:       6608       0.99        <1.0   Unknown
## 40:       6608       0.99        <1.0   Unknown
## 41:       9138       0.09        0.09       G/L
## 42:       9138       0.09        0.09       G/L
## 43:       9219          6         6.0   Unknown
## 44:       9219          6         6.0   Unknown
## 45:      60110       2.16        2.16    mmol/L
## 46:      60110       2.16        2.16    mmol/L
## 47:       6608       0.96        <1.0   Unknown
## 48:       6608       0.96        <1.0   Unknown
## 49:       6608       0.96        <1.0   Unknown
## 50:       6608       0.96        <1.0   Unknown
##     method_num num_result text_result      unit
```

There are 25 duplicate rows identified.

1.5 Save any initial inspection results

```r
# Save the lab data with English column names
write.csv(lab_data, "output_files/lab_data_eng_cols.csv", row.names = FALSE)

# Save column names in a CSV file
# write.csv(names(lab_data), "lab_data_column_names.csv", row.names = FALSE)

# Save unique_test_abbreviations in a CSV file
write.csv(unique_abbr_df, "output_files/unique_test_abbreviations.csv", row.names = FALSE)

# Save unique_test_names in a CSV file
write.csv(unique_names_df, "output_files/unique_test_names.csv", row.names = FALSE)

# Save unique test_name and test_abbr pairs to a CSV file
write.csv(unique_test_pairs_sorted, "output_files/unique_test_name_abbr_pairs.csv", row.names = FALSE)
```

Data set Summary (Final Overview):

*The **lab data** dataset contains **20,487,004 observations** and **8 variables**, with a file size of **1.0 GB**. Most columns have **0 missing values**, but there are some exceptions:*

- *The `test_abbr` column has **387,136 missing entries**.*
- *The `text_result` column has **39 missing values**.*

Additionally:

- ***25 duplicate rows** were identified within the data set.*
- *The data set includes **182,238 unique patients** and **311,611 unique case identifiers**.*
- *There are **3,470 unique test names** and **4,609 unique test abbreviations**.*

Variable Descriptions:

Variable Name	Data Type	Description
`pseudo_patient_id`	Integer	Unique identifier for each patient.
`pseudo_case_id`	Integer	Unique identifier for each case.
`test_name`	Character	Name of the lab test.
`test_abbr`	Character	Abbreviation of the lab test (if available).
`method_num`	Integer	Numerical identifier for the testing method.
`num_result`	Character	Numeric result of the lab test.
`text_result`	Character	Textual representation of the test result.
`unit`	Character	Unit of measurement for the result.